CS-3009 - Deliverable # 03
Faculty Feedback and Evaluation System

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Introduction:

We're building a desktop JavaFX application for faculty feedback and evaluation. This time-period served as our **three development Sprints**, each weeklong. In the first deliverable, we finalized the system requirements alongside the initial product backlog. In second deliverable almost half the implementation of the user stories was done.

For the first Sprint, the goal was to **implement the admin-side dashboard**, since the outputs from admin operations (like registering a user) become the inputs for the rest of the application (e.g. *registered* users can provide feedback to *registered* faculty, which the *registered* heads of departments can review). In the first sprint, we managed to finish the user-part of the admin dashboard.

In the second sprint, we **refined the database schema for the inclusion of the course constraint** (a student can only provide feedback to a teacher for the *course they are enrolled in*). With that, we extended to assign faculty to *registered* courses and enroll students in those courses, while simultaneously working on the **student-side of the panel**.

For the third and the final sprint, the goal was to complete all user stories and start white box and black box testing of the system. We refined the database schema for the inclusion of the department constraint (a CS department HOD can only see the performance scores of its own department). The HOD dashboard is completed, while simultaneously new feedback creation, adding submit feedback functionality at the student side has been done. Also, we extended the admin dashboard to include the log tab so that who accessed the system and for how much time is visible to the admin.

Project vision:

This system is intended for use within students and faculty alike to improve the way feedback is given on the current student portal. The goal is to **encourage transparency and improvement in academic quality**. With this system, a student can confidently provide feedback to the faculty, anonymously, while knowing that the concern would reach the heads as well.

Not only this, but unlike traditional systems, the feedback, though with anonymous authors, will be open to viewing for the rest of the students as well. This can help the students in **understanding the experiences of the previous batches of students** that studied under the same faculty. This would not

only help the students in adapting their study efforts according to the faculty's teaching styles, but would encourage accountability with the faculty heads looking at each student's feedback.

Intended use of the system:

The system will mainly be used by students, faculty members, administrators, and heads of departments.

1. Students will use the system to submit feedback for the courses they are enrolled in, as well as view feedback shared by other students.
2. Faculty members will receive this feedback in an anonymized form, allowing them to understand student concerns and improve their teaching methods.
3. Administrators will manage the overall system, including registering users, assigning faculty to courses, and enrolling students.
4. Heads of Departments will use the system to review feedback across courses and identify patterns, helping them make informed decisions regarding faculty's performance and quality.

During Sprint 3, the stakeholder needs were further refined with the introduction of department-based constraints. It was decided that **HOD should only be viewing performance and health (average score on different performance factor irrespective of the teacher or subject) of its own department only to avoid cross department noise**. This helps in improving the decision taking ability of the HOD purely based on his own data insights and improves the overall reliability of the system.

Features functionality:

The features implemented in each of the two Sprints are clearly described below:

Sprint 1:

- Conducted a timeboxed spike to learn the basics of JavaFX.
- Implemented simple prototype screens (button handling, input fields, basic interaction)
- Developed the initial structure of the **admin dashboard**
- Implemented the core admin functionalities related to user management.

Sprint 2:

- Refined the database schema to include **course-based constraints**
- Ensured that students can only submit feedback for courses they are enrolled in
- Added functionality to **assign faculty to courses**
- Added functionality to **enroll students in courses**
- Added tables to **view the list of students, faculty, and heads**.
- Added tables to **view the list of courses, enrollments and faculty assignments**.
- Added options to view or **modify the user and course data** directly from within the app. ● Began development of the **student-side panel**

Sprint 3:

- Refined the database schema to include **department-based constrains**.
- Ensured that student can submit feedback response for the love forms relevant to their courses and only submit response for the feedback form once.
- Added functionality to add new question of different types (Likert, Slider, Text, Binary) targeting different performance factors (Teaching quality, Content Quality, Punctuality, …).
- Added functionality to create new questionaries/forms by selecting the questions from the existing questions.
- Added functionality to schedule feedback form for specific department and course.
- Added functionality to log the user session login and logout for audit purposes as required by admin.
- For the teacher dashboard added functionality for teacher to view their average scores among different performance factors.
- For the teacher dashboard added functionality for teacher among multiple semesters.
- For the HOD dashboard added the department health tab where HOD can view the average score across different performance factors for analytical decisions and implementation of policies.
- For the HOD dashboard added instructor analysis where HOD can view the top performing teachers for course assignment decisions.
- For the HOD dashboard added instructor analysis where HOD can compare the scores of the teachers of same subject for analytical decisions.
- Done black box testing using techniques like Equivalence Partitioning, Boundary Value Analysis, Error Guessing and prepared the bug report.

User Stories:

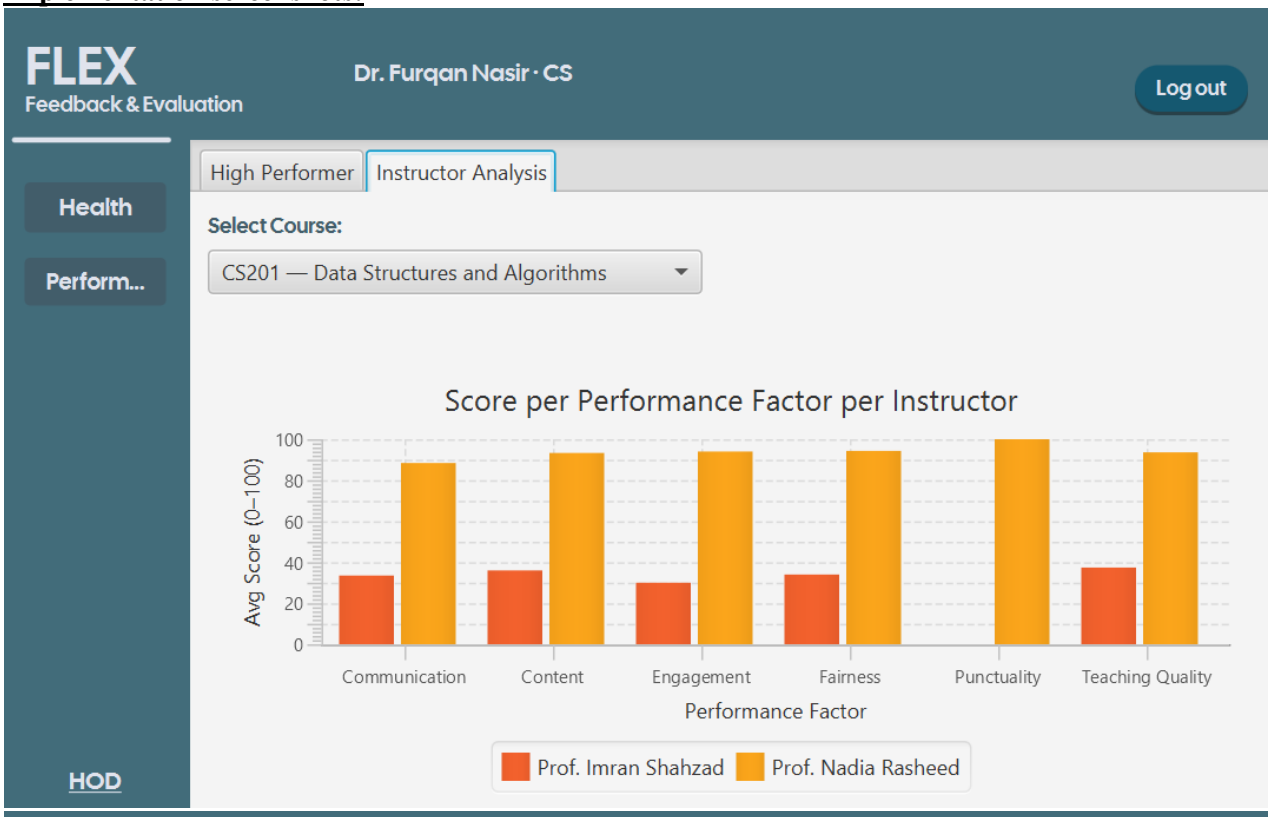
1. As an admin, I want to view the history of which user accessed the system and for how much time.
2. As a student, I want to respond on the live feedback form for reporting my concerns with the teacher and subject.
3. As an admin, I want to add new questions to our question bank to be used in feedback forms.
4. As an admin, I want to add and schedule new feedback forms.
5. As an HOD, I want to view performance graph for overall department health and instructor based growth report.
6. As a teacher, I want to view my scores and track them across several semesters.

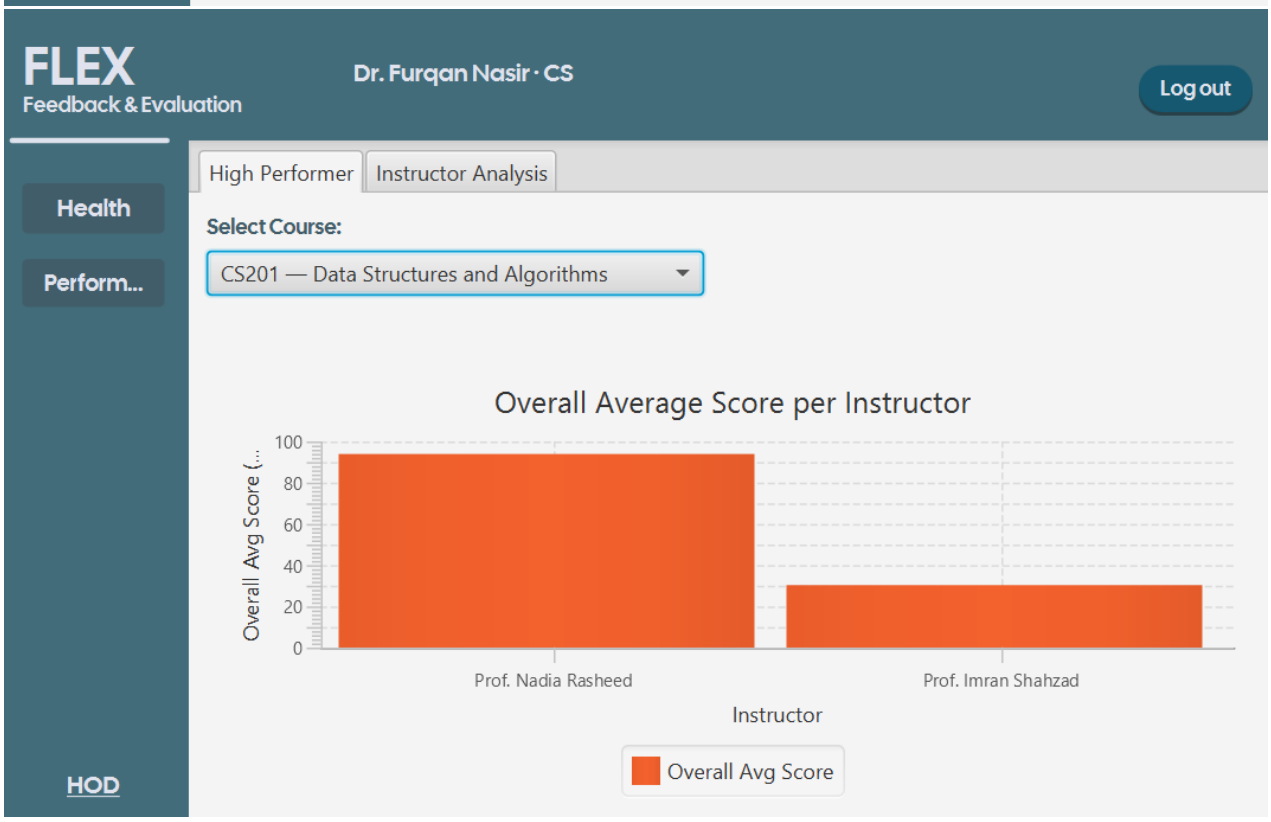
Non-Functional Requirements (NFRs):

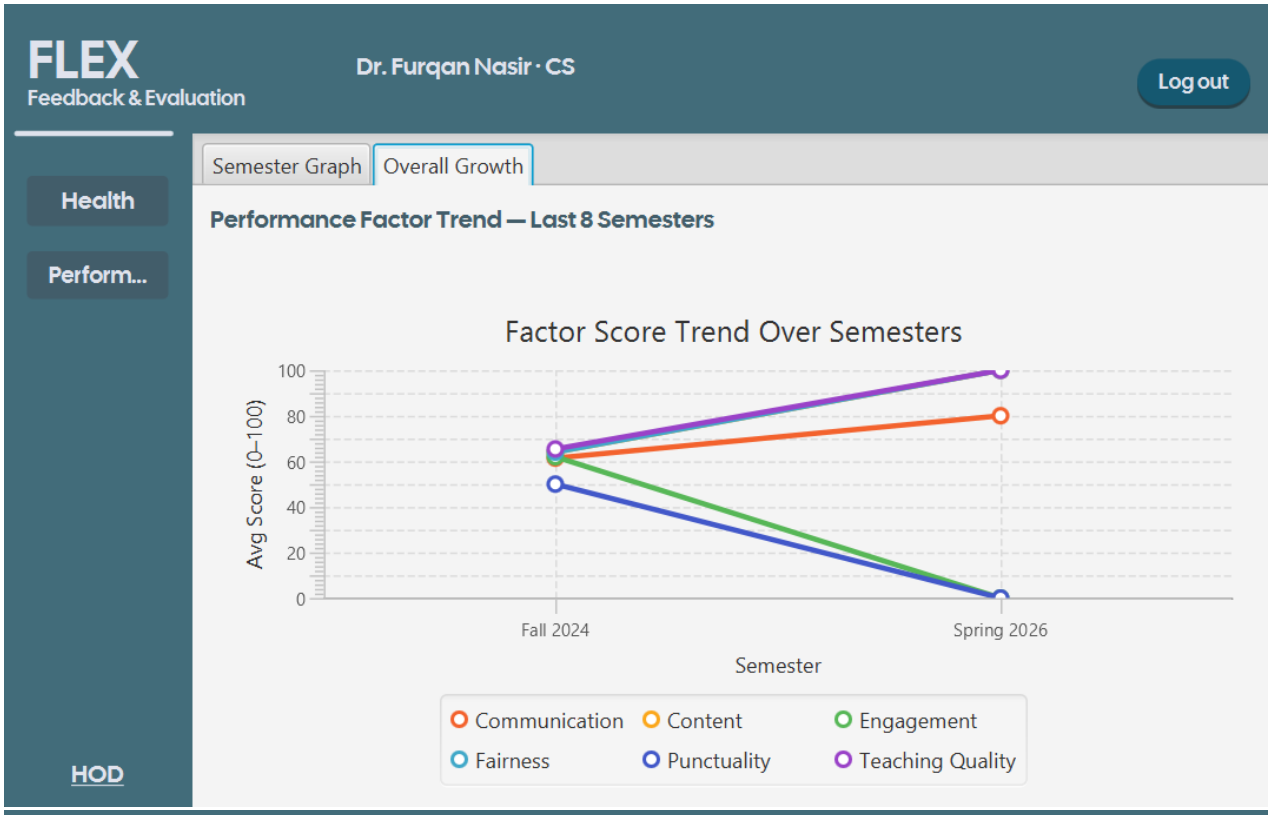
1. The system should respond quickly, with actions like submitting feedback or loading dashboards taking no more than 2 seconds.
2. The interface should be simple and easy to use, so students and faculty can use it without help.
3. Feedback and data must be stored reliably, and the system should handle small errors without losing information.

4. Student feedback must stay anonymous, and only authorized users should access admin functions (via the admin role).
5. The system should handle more users, courses, and feedback over time without slowing down.

Implementation screenshots:






FLEX
Feedback & Evaluation
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Log out
Health
Perform...
HOD
Semester Graph
Overall Growth
Performance Factor Trend — Last 8 Semesters
Factor Score Trend Over Semesters
Avg Score (0–100)
100
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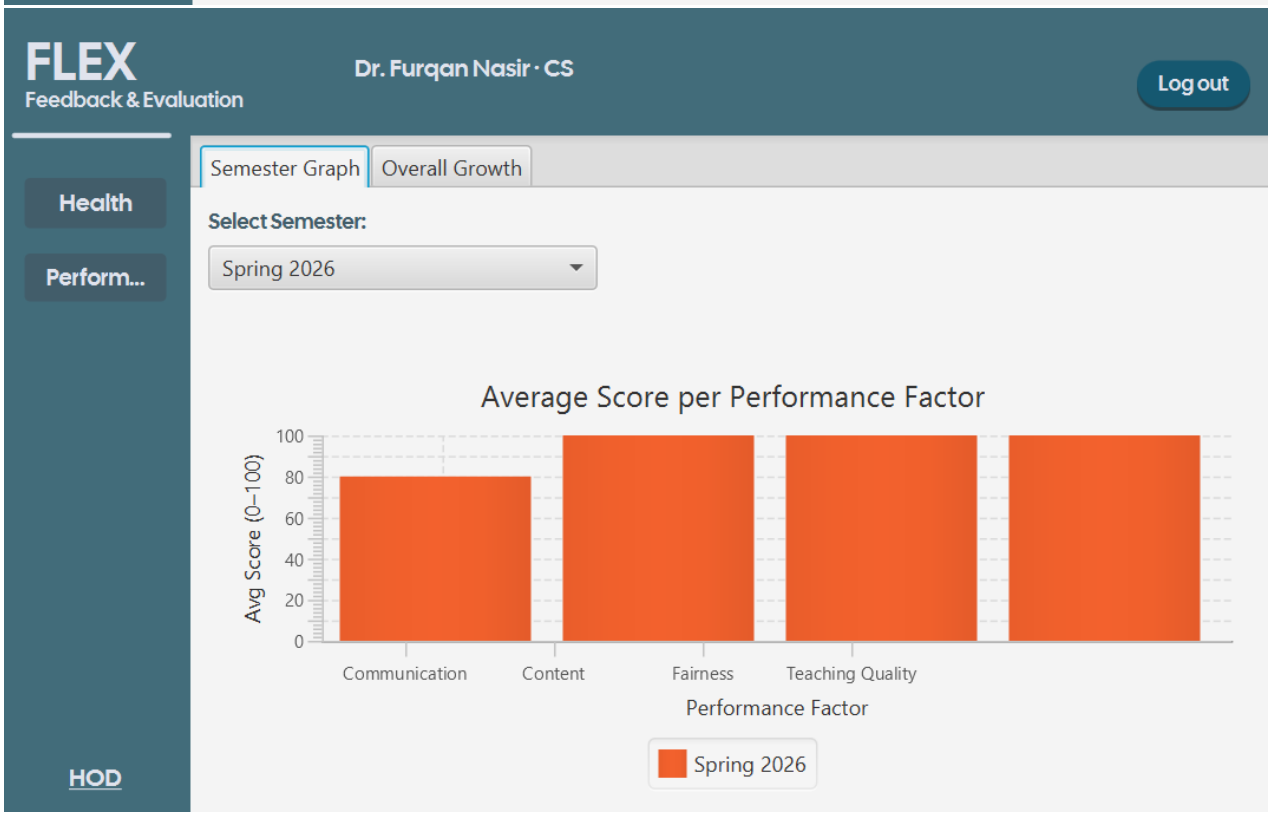
Fall 2024
Spring 2026
Semester
Communication
Content
Engagement
Fairness
Punctuality
Teaching Quality


FLEX
Feedback & Evaluation
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Log out
Health
Perform...
HOD
Semester Graph
Overall Growth
Select Semester:
Spring 2026
Average Score per Performance Factor
Avg Score (0–100)
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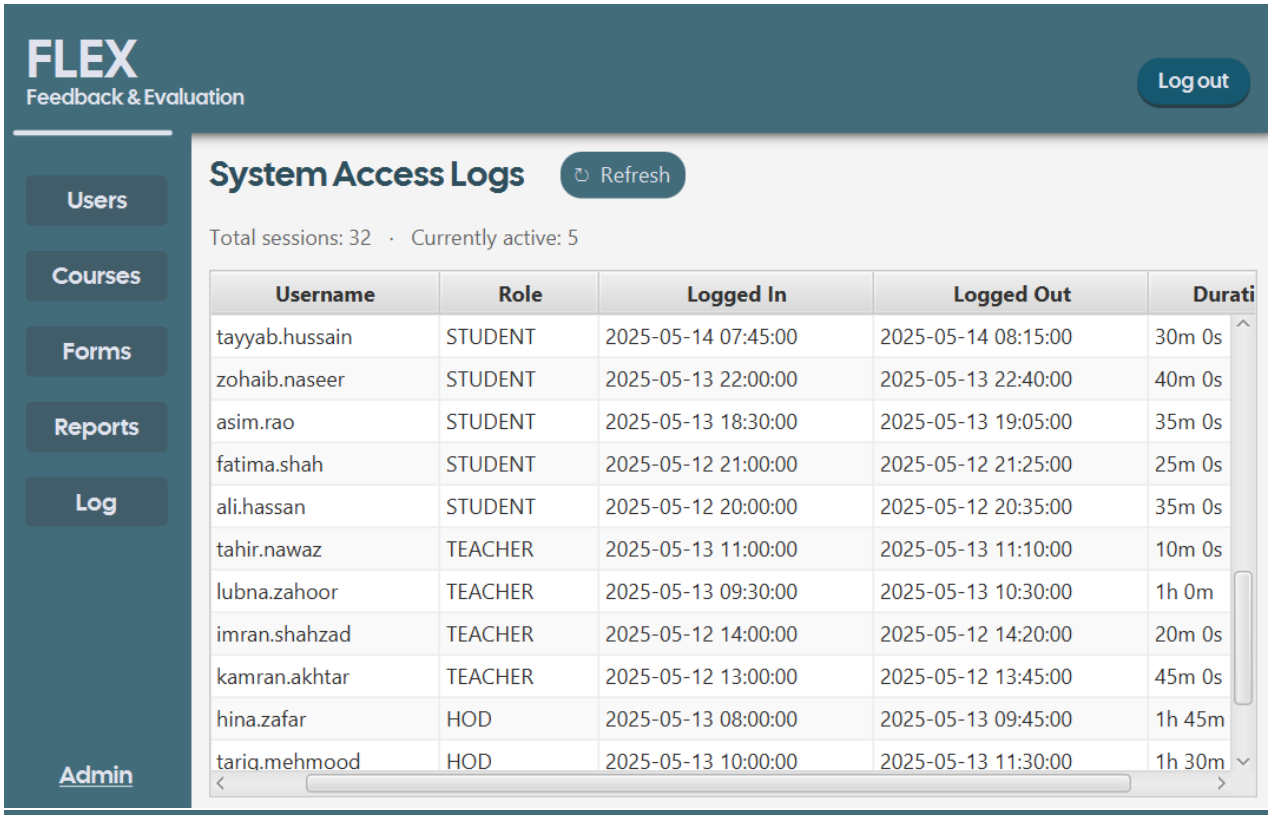
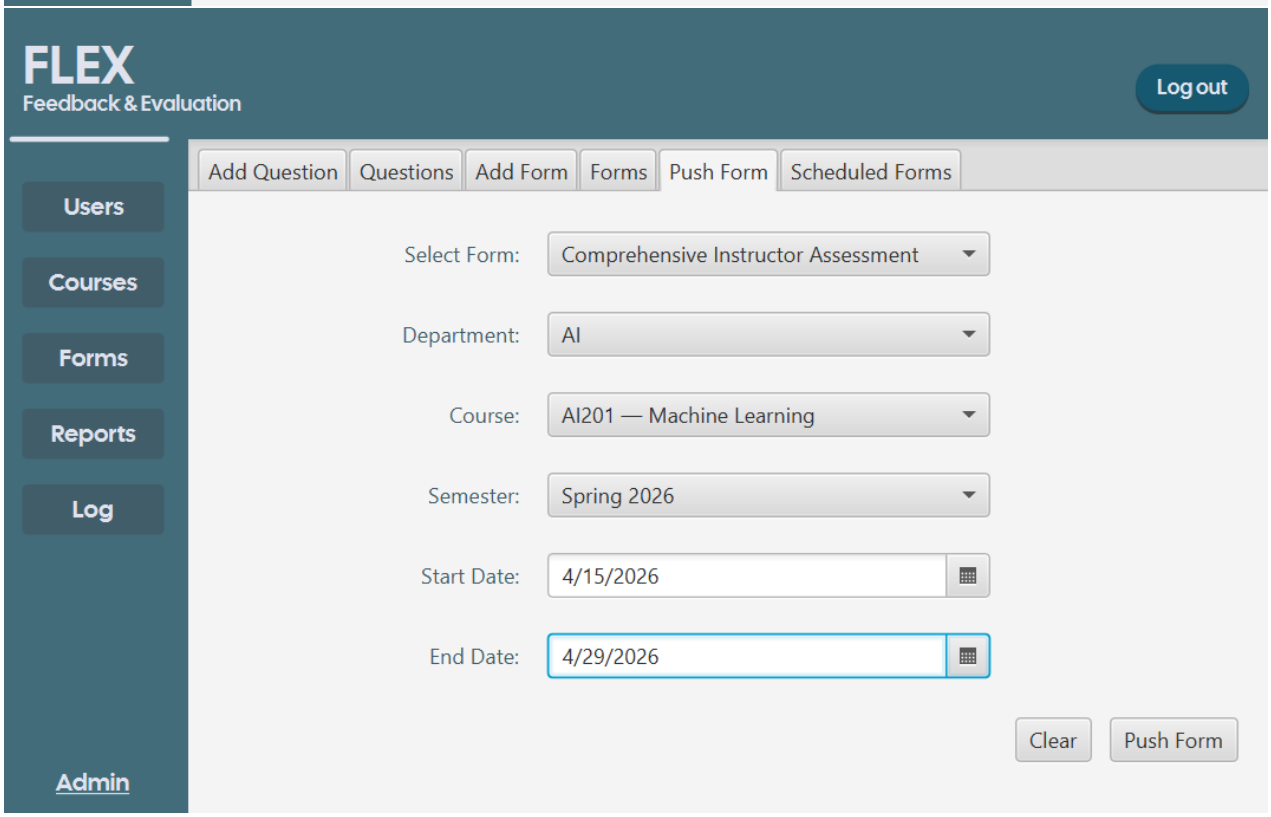
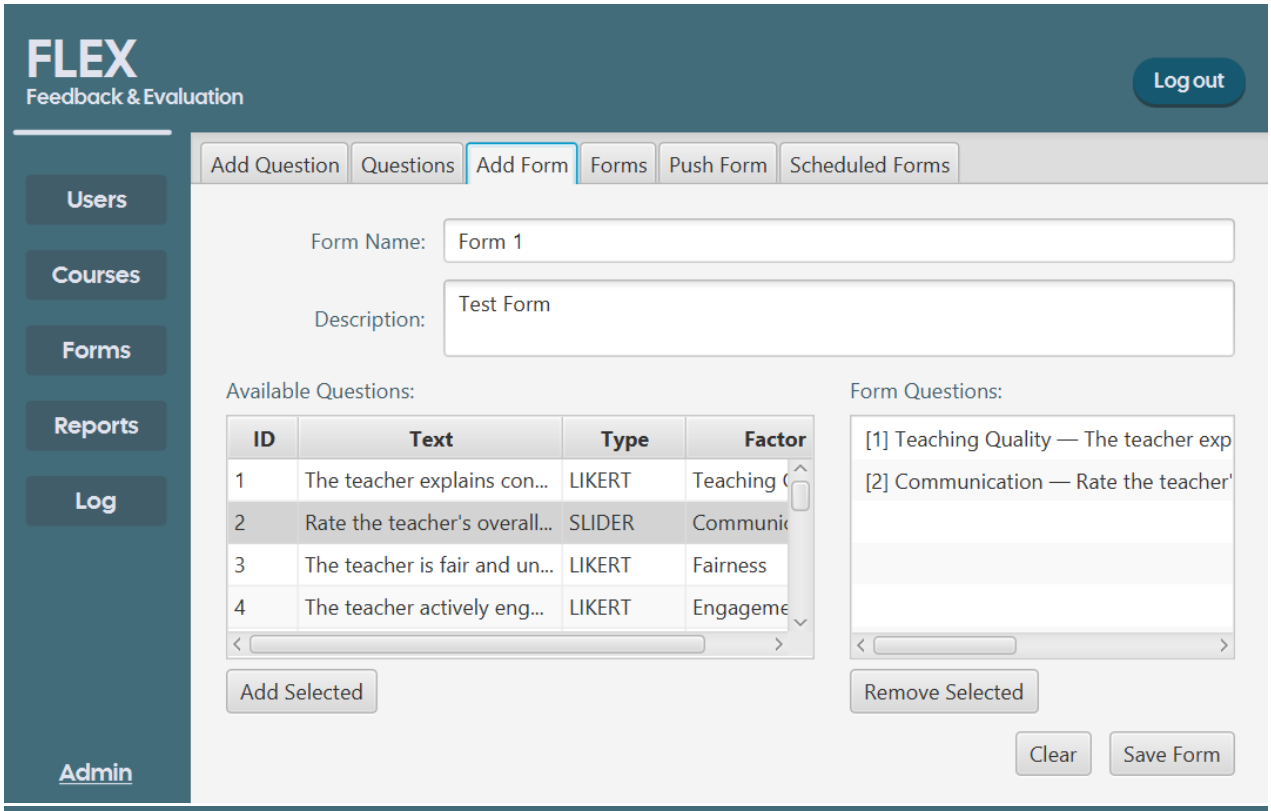
Communication
Content
Fairness
Teaching Quality
Performance Factor
Spring 2026



Username	Role	Logged In	Logged Out	Durati
tayyab.hussain	STUDENT	2025-05-14 07:45:00	2025-05-14 08:15:00	30m 0s
zohaib.naseer	STUDENT	2025-05-13 22:00:00	2025-05-13 22:40:00	40m 0s
asim.rao	STUDENT	2025-05-13 18:30:00	2025-05-13 19:05:00	35m 0s
fatima.shah	STUDENT	2025-05-12 21:00:00	2025-05-12 21:25:00	25m 0s
ali.hassan	STUDENT	2025-05-12 20:00:00	2025-05-12 20:35:00	35m 0s
tahir.nawaz	TEACHER	2025-05-13 11:00:00	2025-05-13 11:10:00	10m 0s
lubna.zahoor	TEACHER	2025-05-13 09:30:00	2025-05-13 10:30:00	1h 0m
imran.shahzad	TEACHER	2025-05-12 14:00:00	2025-05-12 14:20:00	20m 0s
kamran.akhtar	TEACHER	2025-05-12 13:00:00	2025-05-12 13:45:00	45m 0s
hina.zafar	HOD	2025-05-13 08:00:00	2025-05-13 09:45:00	1h 45m
tariq.mehmood	HOD	2025-05-13 10:00:00	2025-05-13 11:30:00	1h 30m

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Feedback & Evaluation
Log out
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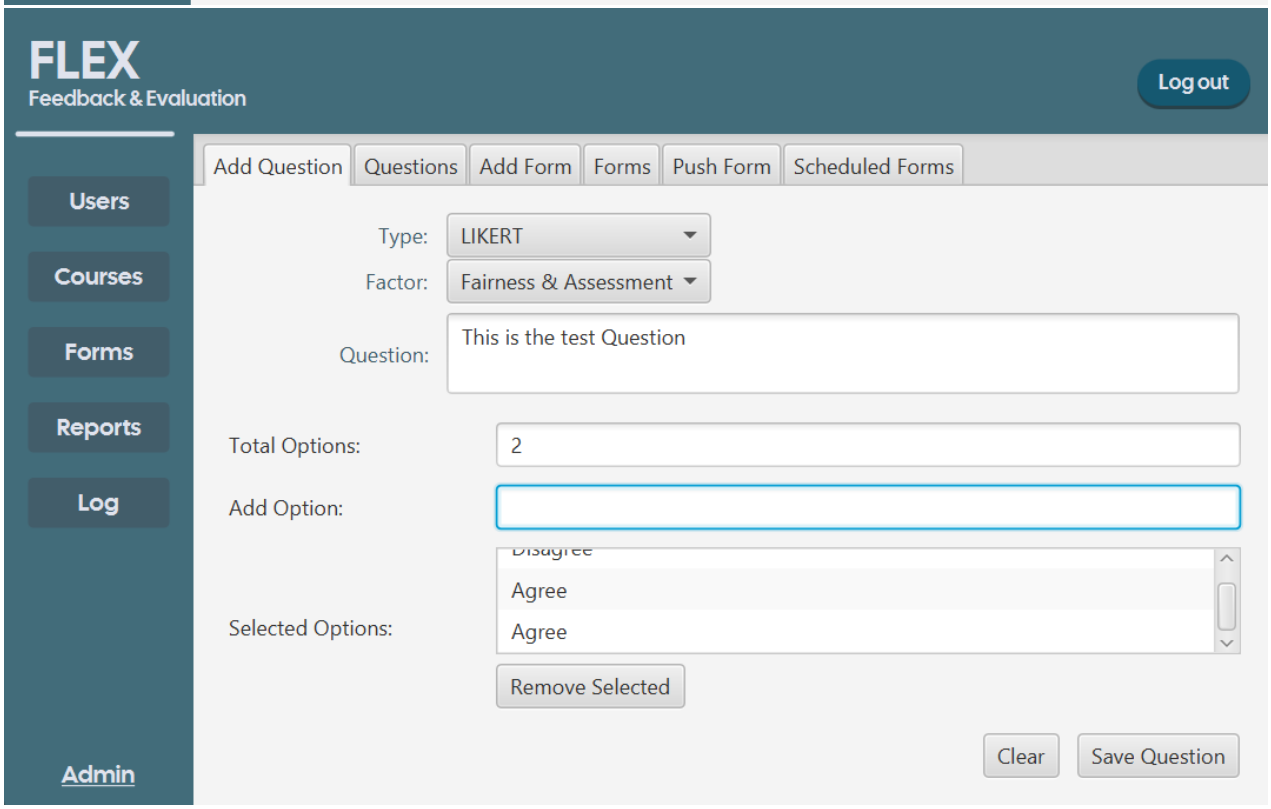
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Fairness & Assessment
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2
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Agree
Agree
Selected Options:
Remove Selected
Clear
Save Question